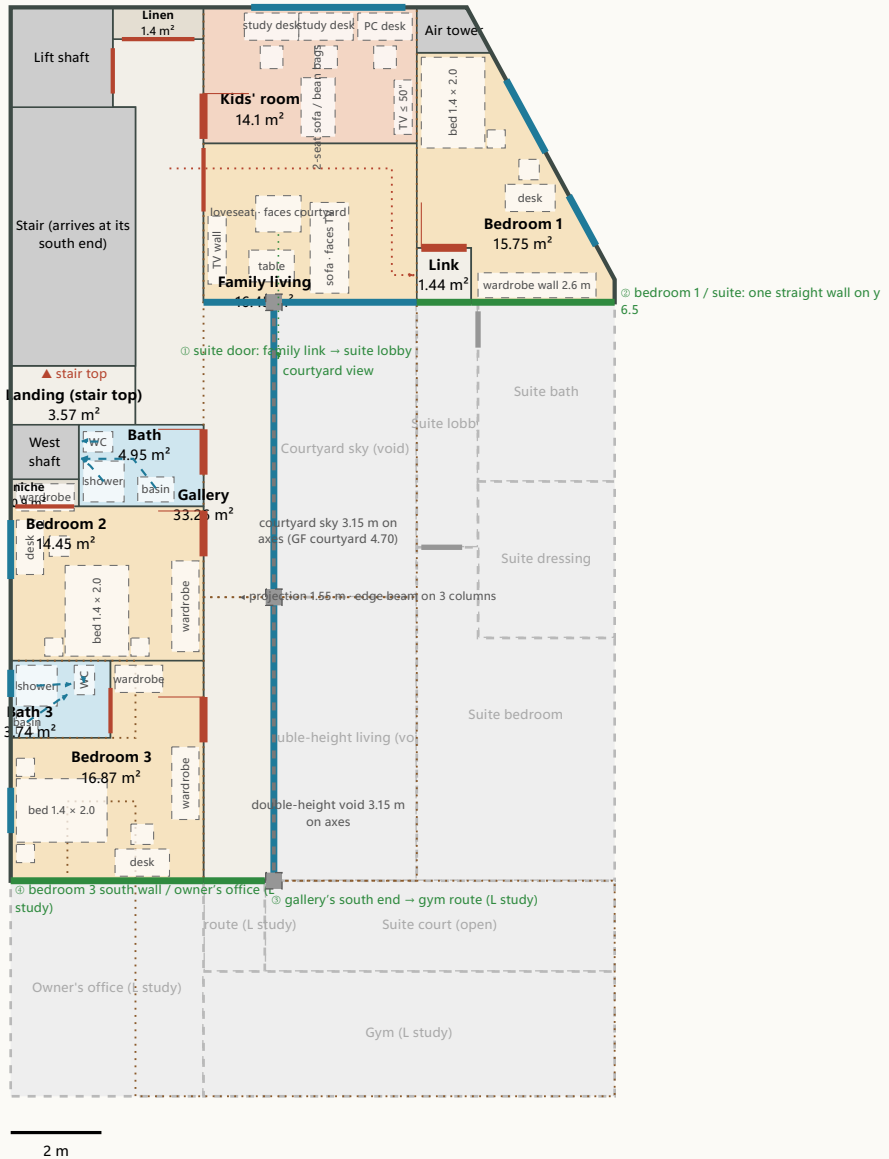
First floor · family part · focus study

Taif / Palm House · first floor V6.3 · local = main CSV coordinates (x east, y south) · wall centre lines · north up

First floor V7.0 · part 1 of 2: family part · west projection 1.55 m on three columns, the gallery and the family living glazed to the courtyard (W1, W2), west rooms 4.25 m wide · part 2 (office, gym, suite court, master suite): studies/ff_suite.typ



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	2.75 × 1.3	3.57	3.14
Gallery (north part + on the projection)	polygon	33.26	30.04
Linen cupboard	2 × 0.7	1.4	–
Family bath (bed 1 + 2, family living)	2.75 × 1.8	4.95	4.04
Wardrobe niche (bed 2)	1.5 × 0.6	0.9	–
Bedroom 2	4.25 × 3.4	14.45	12.84
Bedroom 3 bath (en-suite)	2.2 × 1.7	3.74	2.92
Bedroom 3	polygon	16.87	14.97
Kids' room (study + gaming)	4.7 × 3	14.1	12.44
Family living	4.7 × 3.5	16.45	15.16
Family link	1.2 × 1.2	1.44	1.08
Bedroom 1	polygon	15.75	13.83
Fixed: stair, lift, west shaft, air tower		23.8	
Zone		**150.69**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, ≈ 50 mm balustrade on the void edges, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear · every room has a door reached from circulation · routes ≥ 1.0 m clear · suite door and gallery end on circulation

Key numbers

- **West projection:** 1.55 m over the courtyard and the double-height void (x 4.25–5.8, y 6.5–19.25) on an **edge beam and 3 columns** at x 5.8 the first-floor west wall stands on the GF west wall. Stair, lift, landing and west shaft rise from the ground floor.
- **Courtyard sky** and **double-height void** 3.15 m on axes (the GF courtyard is 4.70); the ground-floor courtyard and living keep their full floor.
- **West rooms** 4.25 m on axes. **Gallery on the projection:** 1.55 m (½ wall + ½ railing off); wider at the stair top. **Circulation in the zone:** 38.28 m² (landing, gallery, family link); the walk through the living is not counted.
- **Family living:** 16.45 m², on the courtyard edge, no separate route. **Kids' room:** 14.1 m².
- **Bedrooms:** 2 · 14.45 m² · 3 · 16.87 m² + its own bath 3.74 m² · 1 · 15.75 m² (no bath). **Family bath** 4.95 m² for bedrooms 1 and 2, the family living and the kids' room.
- **Walked to the family bath door:** bedroom 2 ≈ 2.7 m · family living ≈ 8.8 m · bedroom 1 ≈ 17.3 m. Bedroom 3 → its own bath ≈ 2.7 m.

Plan: red = doors (thin = opening without a leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below (the full courtyard and living floors) · grey dashed = the edge beam (x 5.6) and the three ground-floor columns · grey = neighbours · dashed boxes = furniture and fixtures.

Family part · the west projection and the rooms

The west projection · 1.55 m over the courtyard and the double-height void

- **Where:** along the west edge of the courtyard and of the double-height living, x 4.25–5.8, from the courtyard's north edge (y 6.5) to the living's south wall (y 19.25). The west rooms take the full band over the GF west band and riwaq; their west wall stands on the GF west wall.
- **Widths:** rooms 4.25 m on axes (≥ 4.0 m clear), gallery 1.55 m (≥ 1.0 m clear).
- **Carried on:** an edge beam along x 5.8 and 3 columns, each in an existing wall or glass line, none free-standing in a room: column 1 in the women's majlis south wall (y 6.5), column 2 a pier in the glass between the courtyard and the living (y 13.0), column 3 in the living's south wall (y 19.25). On the ground floor it reads as a colonnade along the courtyard edge.
- **Why columns and not a cantilever wall:** walls stand directly on walls; the earlier split projection (a west wall on cantilever beams, 0.11 m from the M line) was reverted.
- **Gallery:** closed to the courtyard by the steel window W1 on the upstand of the edge beam (sill +3.82, guarding transom +4.70, head +6.75, four inward casements; V6.4, enclosure option A), wider at the stair top; every west room door opens from it; no room opens onto the stair landing. Clear 1.405 m (1.235 m at the three column sleeves Ø0.40).

Kids' room · family living · bedroom 1

- **Kids' room** 14.1 m^2: two study desks + PC desk under the north window, a gaming corner (TV on the east wall), solid door from the gallery.
- **Family living** 16.45 m^2: a view, not a path; the walk to the family link passes behind and beside the sofa.
- **Bedroom 1:** one room, no bath, its south wall straight on y 6.5; it uses the family bath.

Family bath · 2.75 × 1.8 m, beside the west shaft

- WC on the shaft wall, shower, basin by the door; door from the gallery (opening in, swing clear), not onto the landing. For bedrooms 1 and 2, the family living and the kids' room. Its west wall is the shaft (x 1.5), its east wall the band's gallery wall.

Bedroom 2 · 4.25 × 3.4 m + wardrobe niche 1.5 m

- **Bed 1.4 × 2.0** with its head on the south wall and bedside tables on both sides; **wardrobes** in the niche beside the shaft and on the east wall; **desk** under the west window. Door from the gallery (swing clear), ≈ 2.7 m from the family bath.

Bedroom 3 · 16.87 m^2 + its own bath 3.74 m^2

- An entry with a wardrobe, then the room: **bed 1.4 × 2.0** head on the west wall with two bedside tables, wardrobe on the east wall, desk on the south wall. Door from the gallery (swing clear).
- **Own bath (en-suite)** off its entry: shower, WC, basin; ≈ 2.7 m from the bedroom door. It stands **exactly over the GF family WC** (walls on the GF lines y 14.4 and 16.1).

Gym route and the owner's office → the L study

- The gallery ends at y 19.25; from there the **gym route** and the **owner's office** belong to the L study (studies/ff_suite.typ), with the gym, the court and the suite.

Interfaces with the L study (fixed)

- ① suite door at the end of the family link (y 6.5); ② bedroom 1 / suite: one straight wall on y 6.5; ③ the gallery's south end (y 19.25), where the gym route starts; ④ bedroom 3's south wall (y 19.25), a wall to the owner's office (no window).

Family part · routes, walls, drains, open items

Routes (≥ 1.0 m clear)

Route	Axes	≈ clear	Deducted
Landing	1.3	1.2	½ wall on the south side; the north side is the stair arrival
Gallery north (beside the stair)	1.5	1.3	½ wall each side
Gallery on the projection	1.55	1.42	½ wall (rooms side) + ½ window frame W1 on the B3 upstand (void side)
Walk behind the sofa (in the living)	1.1	1	½ wall on the kids' room side; the sofa side is open
Walk beside the sofa to the family link	1.1	1	½ wall on the bedroom 1 side; the sofa side is open
Family link	1.2	1	½ wall each side (bedroom 1 to the north and east)

Walking distance	m
Stair top → kids' room door	8.8
Stair top → family living (opening)	8.7
Stair top → bedroom 1 door	15.9
Stair top → ① suite door	15.9
Stair top → family bath door	5.7
Stair top → bedroom 2 door	7.5
Stair top → bedroom 3 door	10.9
Stair top → gallery's south end (L)	14.5
Bedroom 2 door → family bath door	2.7
Family living → family bath door	8.8
Bedroom 3 door → its own bath door	2.7
Bedroom 1 door → family bath door	17.3

Stair top → every room, the suite door and the gallery's south end runs only on the landing, the gallery, the family link and the walking line through the living (checked: every door's approach is on circulation, no furniture in a route).

Engineer

- **Projection:** 1.55 m slab beyond x 4.25 along y 6.5–19.25; an **edge beam** on x 5.8 (spans ≈ 6.5 m and 6.25 m, depth to confirm; seen from the courtyard and the living, expressed as the colonnade's beam); 3 **columns** down to their own foundations (sizes and pads to confirm).
- **Ground floor** changes only by the columns, their foundations, the beam line and the lower ceiling under the projection.

Walls off the ground-floor lines

Ground-floor lines here: x 0, 1.25 (guest WC), 1.5 (shaft), 2.25, 2.75, 4.25, 8.95 · y 2.2, 7.9, 10.4, 12.9, 14.4, 16.1, 17.5, 19.25, 21.4 (foyer) (west band); the women's majlis (x 4.25–8.95, y 0–6.5) and the men's majlis (x 2.75–13.311, y 19.25–24) have no inner wall; east part (bedroom 1, family link): over the GF maid wing, laundry and the kitchen's north wall.

- **Off:** y 3.0 (kids' room / living, over the women's majlis); y 11.0 (family bath and niche / bedroom 2) and x 1.5 south of the shaft (niche); x 2.2 (bedroom 3 bath / entry).
- **On:** y 14.4 and 16.1 (bedroom 3 bath = the GF family WC walls), x 1.5 / y 10.4 (shaft), y 6.5 (bedroom 1 / suite), **x 4.25** (the rooms' east wall, on the GF courtyard edge line), the outer walls.
- The GF riwaq wall x 2.75 carries the floor; the rooms span over the riwaq.

Drains (blue dashed)

- **Family bath:** WC on the west shaft wall, shower and basin in the bath floor to the shaft; floor over the GF utility room and hospitality kitchen, nothing in their ceilings.
- **Bedroom 3 bath:** exactly over the GF family WC: WC straight into its stack, shower and basin in the floor build-up. Wet room on a wet room.
- **Bedroom 1:** no bath. No east duct (the suite bath drains to a stack outside the east wall, see ff_suite.typ).

Open items (all in docs/OPEN_ITEMS.md)

- **Engineer:** the edge beam and the 3 columns at x 5.8 (sizes and foundations); walls off the GF lines.
- **Gallery window W1** on the B3 upstand and **W2** on the family living's edge (black steel, divided lites, laminated glass below +4.70): the courtyard stays open to the sky, the house is closed round it (V6.4).
- **Built-in cupboards in niches** (linen cupboard, bedroom 2 wardrobe niche): joinery boxes scribed to the walls; decide with the carpenter.
- **Courtyard light:** a daylight study of the courtyard and the double-height living.